A HOSPITAL FINANCE STUDY

The economics of care

What 3,103 hospital cost reports show about utilization, cost structure and ownership

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Why I chose this question

WHERE IT STARTED

How do financial priorities shape staffing and the delivery of care?

A class discussion about private equity in healthcare made me curious about the economic pressures behind those decisions.

Hospital cost reports gave me a way to investigate them with institutional data, one measurable piece at a time.

RESEARCH QUESTION

Which hospital characteristics are associated with operating margin: occupancy, labor cost share, uncompensated care or ownership?

What the data shows

−3.0 pts

Uncompensated care is the largest drag

A 10-point higher uncompensated-care share of costs goes with about a 3.0-point lower margin.

+2.8 pts

Ownership adds explanatory power

Adding ownership type lifts R^2 from 7.0% to 9.8%. Group averages span about 14 points.

0.059

Margins stay hard to predict

On held-out hospitals the OLS model reaches a test R^2 of 0.059. Most variation is left unexplained.

These are associations among 3,103 short-term hospitals in one year of cost reports, not causal effects.

Source: final report, 2 March 2026 · coefficients per 10-point change in each predictor

How I built the analysis

01 Organize

Normalized one cost-report CSV into four PostgreSQL tables (3NF), joined on provider ID.

02 Clean

Traced extreme ratios to tiny denominators, then set a revenue floor and bounded margins.

03 Explore

Compared margin distribution, ownership groups and pairwise correlations in SQL and Python.

04 Model

Fit OLS and Ridge regressions, then tested both on a held-out 20% of hospitals.

TOOLS

PostgreSQL · SQL · Python · pandas · statsmodels · scikit-learn · Matplotlib

Sources: final report and database working notes

A $1 denominator broke the first margin calculation

$1

net patient revenue in an inspected hospital record

−$129.9M

net income in that same record, a margin below −100,000,000%

THE FIX

Keep hospitals with net patient revenue above $1 million, and bound operating margin between −100% and +100%.

Source: database working notes · The exclusions stabilize the ratio and also change which hospitals the results describe

Three sample sizes, each tied to a specific step

3,123

Financial screen

Short-term hospitals with revenue above $1M, non-null net income and a bounded margin.

3,103

Analytic sample

Adds the cost and utilization requirements: positive total costs and available bed-days.

3,033

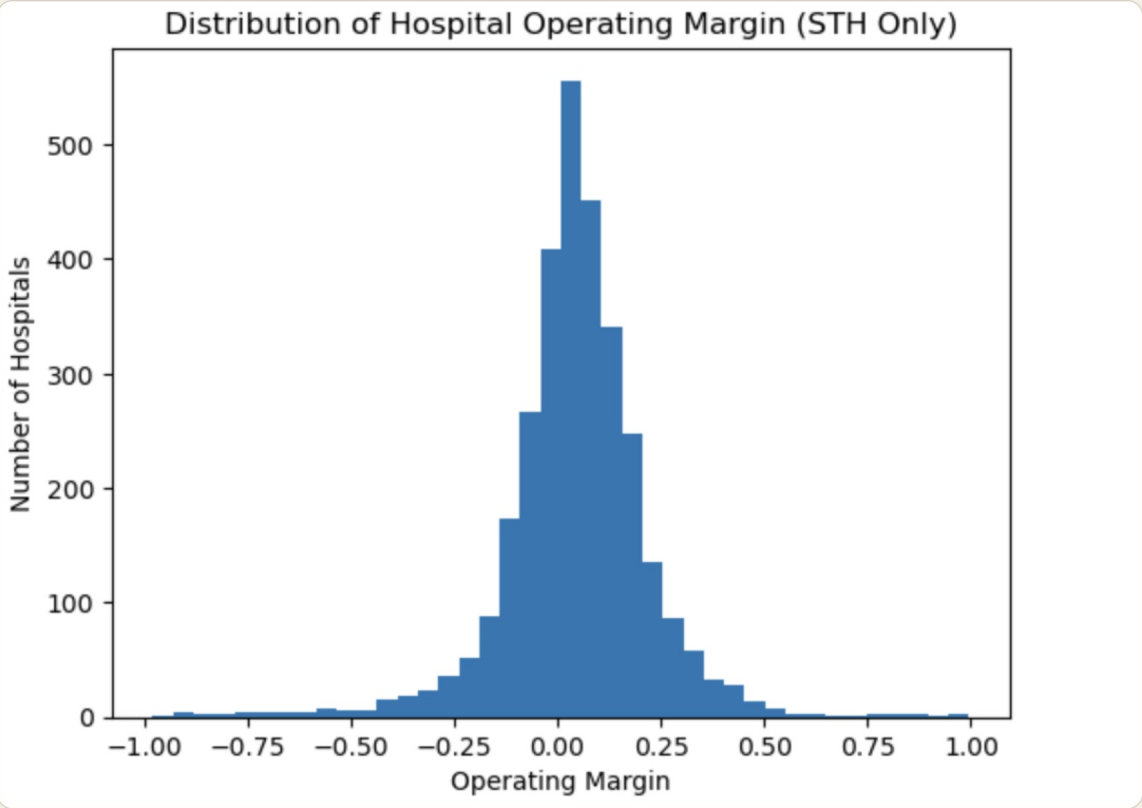
Baseline regression

Observations in the three-predictor OLS. The ownership model used all 3,103.

These counts describe different populations, so every figure in this deck names the sample behind it.

Sources: SQL screening query, final report, OLS regression output

A positive median, with a third of hospitals in the red



4.55%

median operating margin

−3.3% to 13.0%

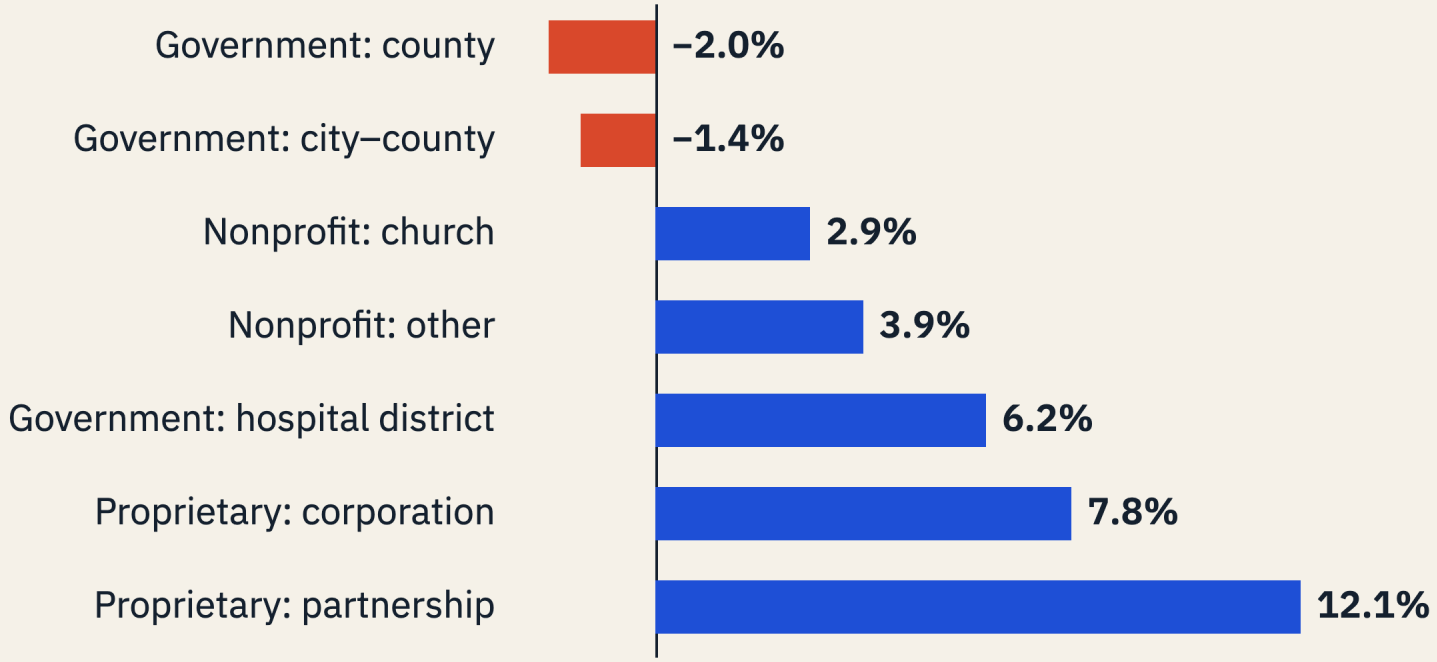
range for the middle half of hospitals

33.8%

had a negative margin: 1,055 of 3,123 screened hospitals

Source: final report histogram and percentiles; loss share from the financial screen (n = 3,123)

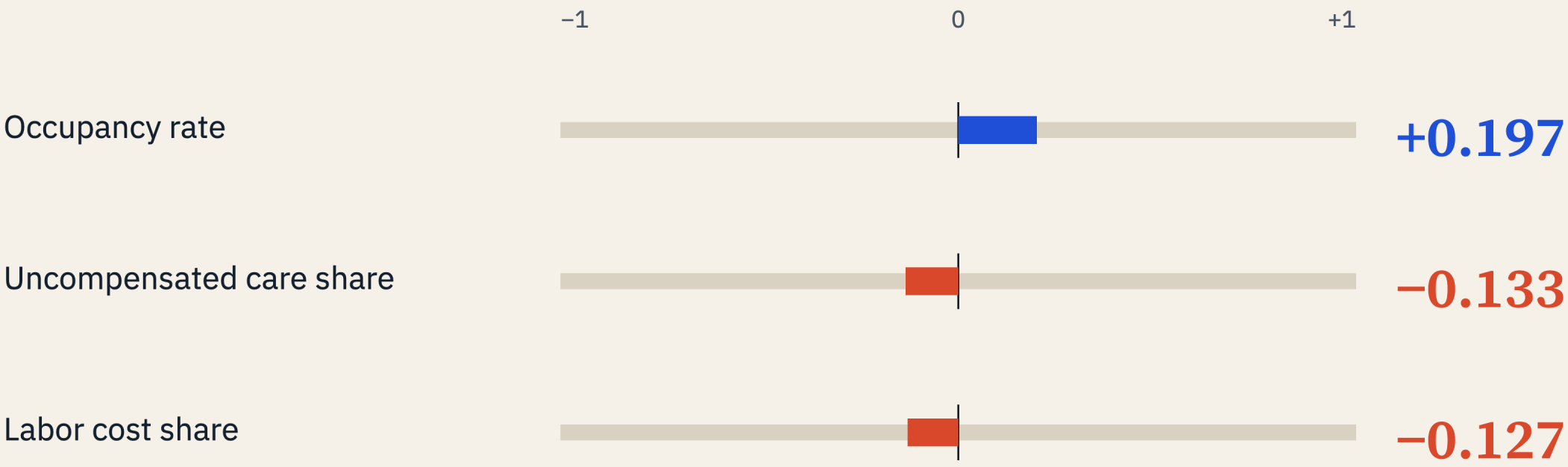
Ownership groups differ by up to 14 points



Average operating margin, selected groups. Partnerships average 12.1% but include only 79 hospitals. The largest group, nonprofit other, has 1,539 hospitals and averages 3.9%.

Source: ownership summary query (type of control) · descriptive averages, not evidence of why the groups differ

Pairwise correlations with margin are weak

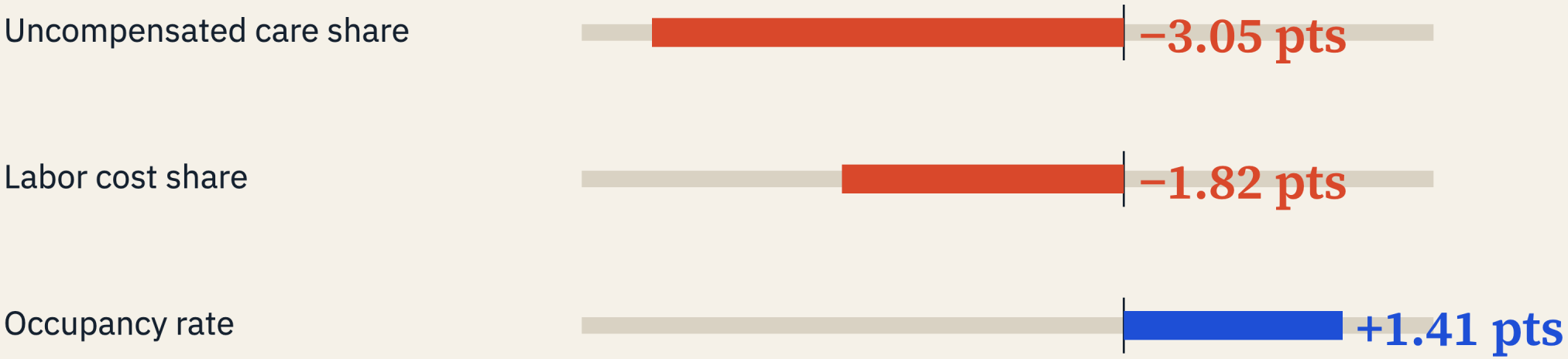


Occupancy has the largest correlation of the three, and it still sits close to zero on the full −1 to +1 scale.

Source: correlation queries, short-term hospitals · Pearson r describes association, not the effect of changing a policy

Uncompensated care has the largest coefficient

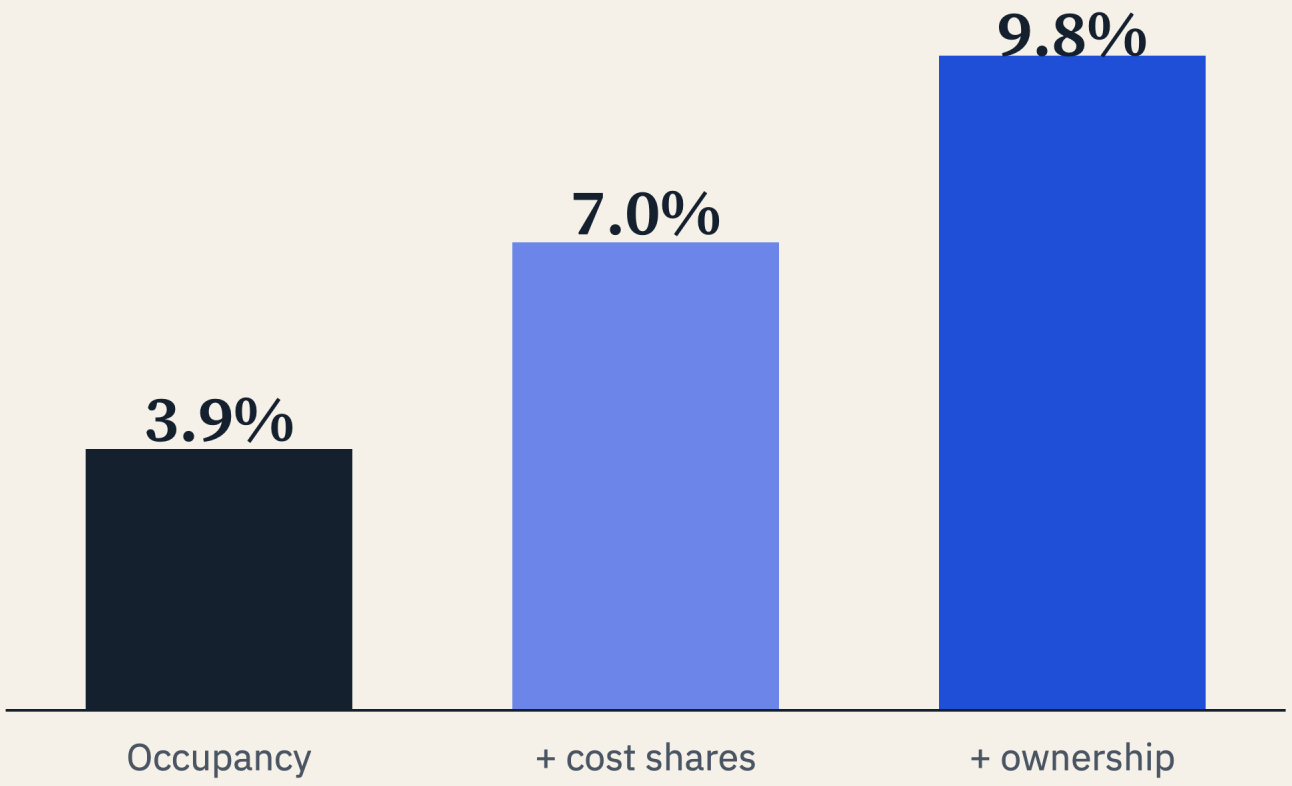
Associated change in operating margin for a 10-point increase in each predictor, holding the other two constant



With ownership controls added, uncompensated care stays the largest effect at −3.0 points per 10 (labor cost share −1.5, occupancy +1.7).

Source: OLS regression, n = 3,033 (baseline) and n = 3,103 (with ownership) · coefficients × 10 · associations only

Explained variation stays under 10%



90.2%

of variation in margin remains unexplained, even with ownership in the model.

Model samples differ slightly (3,033 and 3,103 hospitals), so the increments are approximate.

Source: in-sample R^2 from the final report's OLS models

Prediction is a harder test

OLS · TEST R²

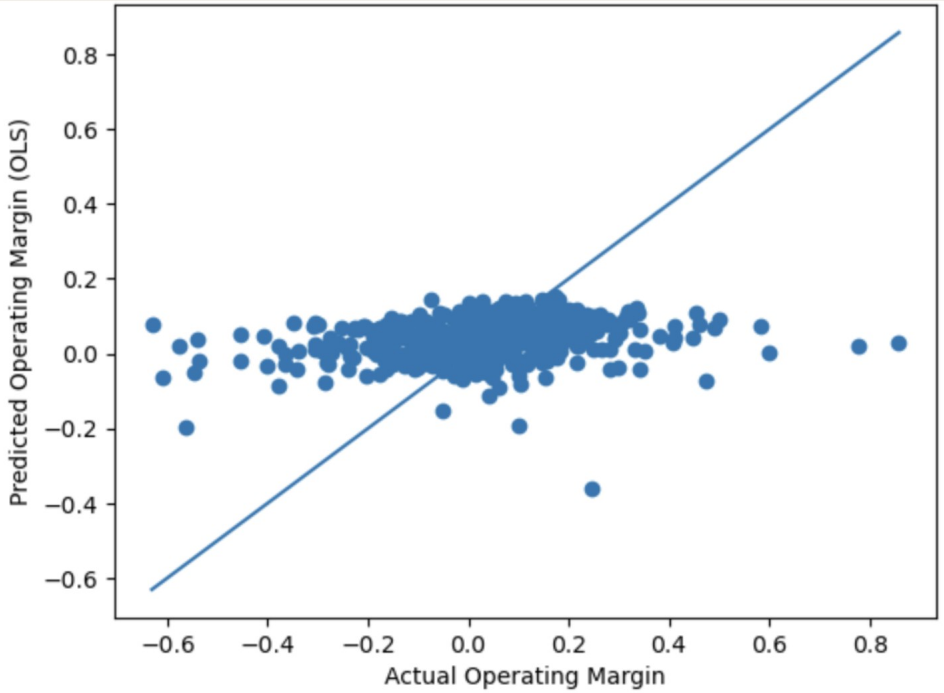
0.0589

Root mean squared error: 16.48 points

RIDGE · TEST R²

0.0407

Root mean squared error: 16.64 points



Predictions cluster near the center while actual outcomes spread far wider. The model captures direction, not individual hospitals.

Source: 80/20 train-test split · OLS performed slightly better than Ridge on held-out data

Where the analysis could go next

CAPACITY

Where is available capacity underused?

Examine demand, throughput and quality constraints before treating occupancy as a target.

COST MIX

What sits behind a high labor share?

Separate staffing need, patient acuity and productivity from the headline cost ratio.

REIMBURSEMENT

Where is uncompensated burden concentrated?

Add payer mix, coverage and collections context to the financial picture.

PEER GROUPS

Which hospitals are truly comparable?

Compare ownership alongside market, service mix and funding structure.

Proposed questions derived from the findings, not completed analysis

What these results can and cannot support

LIMITS

- Associations, not causal effects: one year of cross-sectional data.
- The revenue floor and margin bounds change which hospitals the results describe.
- Explanatory power is low: R^2 between 0.07 and 0.10.
- Ratios inherit the definitions and reporting quality of the cost-report fields.

HOW I WOULD STRENGTHEN IT

- Verify source fields, join grain and row counts at each step.
- Refit every model on one common sample.
- Vary the trimming thresholds and examine influential records.
- Extend the analysis across multiple years.

Source: final report, 2 March 2026

What I brought to the project

SQL AND DATA DESIGN

Organized one CSV into a four-table PostgreSQL schema and built the margin, occupancy, labor and uncompensated-care ratios in SQL.

MODELING

Fit OLS and Ridge regressions in Python, checked residual diagnostics and validated both on a held-out 20% of hospitals.

JUDGMENT

Questioned unstable ratios before interpreting results, and reported modest explanatory power plainly.

Hospital finances need operational context and careful measurement.

Appendix: Definitions and sample

Measure	Calculation
Operating margin	Net income ÷ net patient revenue
Occupancy rate	Patient days, all payers ÷ available bed-days
Labor cost share	Adjusted salaries ÷ total costs
Uncompensated care share	Unreimbursed and uncompensated care ÷ total costs

Analytic sample (n = 3,103): short-term hospitals with net patient revenue above $1M, non-null net income, positive total costs and bed-days, and operating margin between –100% and +100%.

Source: final report SQL definitions · 2023 Medicare hospital cost reports

Appendix: Model results

Model	Predictors	Observations	R^2
Bivariate OLS	Occupancy rate	Not reported	0.039
Multivariate OLS	Occupancy, labor cost share, uncompensated care share	3,033	0.070
OLS with ownership	Same three, plus ownership-type indicators	3,103	0.098

Held-out test set (80/20 split): OLS R^2 0.0589 with RMSE 0.1648; Ridge R^2 0.0407 with RMSE 0.1664. Ridge used a cross-validated penalty. Residuals are heavy-tailed (kurtosis about 8.7), so inference relies on the large sample.

Source: OLS and Ridge output in the final report · adjusted R^2 0.069 and 0.094 for the last two models